18. Two liquids X and Y on mixing gives a warm solution. The solution is

a) ideal

b) non-ideal and shows positive deviation from Raoults law

c) ideal and shows negative deviation from Raoults Law

d) non-ideal and shows negative deviation from Raoults Law

19. The relative lowering of vapour pressure of a sugar solution in water is 3.5×10^{-3}. The mole fraction of water in that solution is

a) 0.0035 b) 0.35

c) 0.0035 / 18 d) 0.9965

20. The mass of a non-voltaile solute (molar mass 80 g mol^{-1}) which should be dissolved in 92g of toluene to reduce its vapour pressure to 90%

a) 10g b) 20g

c) 9.2 g d) 8.89g

21. For a solution, the plot of osmotic pressure (π) verses the concentration (c in mol L^{-1}) gives a straight line with slope 310R where 'R' is the gas constant. The temperature at which osmotic pressure measured is

a) 310×0.082 K b) 310°C

c) 37°C d) $\frac{310}{0.082}$ K

22. 200ml of an aqueous solution of a protein contains 1.26g of protein. At 300K, the osmotic pressure of this solution is found to be 2.52×10^{-3} bar. The molar mass of protein will be (R = 0.083 L bar mol^{-1} K^{-1})

a) 62.22 Kg mol–1 b) 12444g mol^{-1}

c) 300g mol^{-1} d) none of these

23. The Van't Hoff factor (i) for a dilute aqueous solution of the strong elecrolyte barium hydroxide is (NEET)

a) 0 b) 1

c) 2 d) 3

24. What is the molality of a 10% W/W aqueous sodium hydroxide solution ?

a) 2.778 b) 2.5

c) 10 d) 0.4

25. The correct equation for the degree of an associating solute, 'n' molecules of which undergoes association in solution, is

a) $\alpha = \frac{n(i-1)}{n-1}$ b) $\alpha^2 = \frac{n(1-i)}{(n-1)}$

c) $\alpha = \frac{n(i-1)}{1-n}$ d) $\alpha = \frac{n(1-i)}{n(1-i)}$

26. Which of the following aqueous solutions has the highest boiling point ?

a) 0.1M KNO_3 b) 0.1 M Na_3PO_4

c) 0.1 M $BaCl_2$ d) 0.1 M K_2SO_4

27. The freezing point depression constant for water is 1.86° K $Kgmol^{-1}$. If 5g Na_2SO_4 is dissolved in 45g water, the depression in freezing point is 3.64°C. The Vant Hoff factor for Na_2SO_4 is

a) 2.50 b) 2.63

c) 3.64 d) 5.50

28. Equimolal aqueous solutions of NaCl and KCl are prepared. If the freezing point of NaCl is –2°C, the freezing point of KCl solution is expected to be

a) –2°C b) – 4°C

c) – 1°C d) 0°C

Unit 9 - jagan.indd 63 27-01-2021 15:06:32